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BAN402 Decision Modelling in Business

Project 1

Candidate numbers: 39 **and** 115

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Introduction

In this project we seek to solve LP problems using the engine `cplex` in the AMPL IDE. Adding on this written project, we also encourage the reader to follow along in the `.zip`-file named **p2-ampl-files**, with file names corresponding to the sections `A.mod, A.dat, A.mod = ` PART A, `B.mod, B.dat, B.mod = ` PART B etc. We have also added some printouts from AMPL in the *Appendix*.

PART A

Part A Task 1 - Formulate a linear programming model

Under follows a model for deciding the units of intensity at which each abatement method must be used, such that the optimal solution minimizes the cost of reducing emissions. In the model, the phrase "each abatement method must be used" is interpreted as requiring each method's inclusion in the model, rather than mandating a non-zero allocation. Therefore, $x \geq 0$ is used instead of $x \geq 1$.

Sets

M : Set of abatement options $\qquad$ G : Set of greenhouse gases

Parameters

$$\text{reduction}(m, g) : \forall m \in M, g \in G : \text{Emission reduction of gas } g \text{ by abatement option } m$$
$$\text{cost}(m) : \forall m \in M : \text{Cost of abatement option } m$$
$$\text{target}(g) : \forall g \in G : \text{Emission target for gas } g$$

Variables

$$x(m) \geq 0 : \forall m \in M : \text{Units of intensity used for method}$$

Objective

$$\text{Minimize : total\_cost} = \sum_{m \in M} x(m) \times \text{cost}(m)$$

Constraints

$$\text{reduction\_emissions}(g) : \forall g \in G : \sum_{m\ inM} \text{reduction}(m, g) \times x(m) \geq \text{target}(g)$$

Implement the model in AMPL and solve it

SCR, LEF, and EP are M1, M2, and M3 respectively. The optimal solution is to allocate 1.3333 units to SCR, 11.1111 units of intensity to EP, and none to LEF. The optimal gives an objective value, total cost, of \$13,000. All numerical values in Part A are sourced from the `A.txt` file, generated by executing the `A.run` file. The complete output is provided in the Appendix for reference[1].

Part A Task 2 - Constraints in optimal solution

In our optimization model, we incorporate a constraint labeled as `reduction_emissions`. Currently, the values for generators G1, G2, and G3 are 140, 121.778, and 180 respectively. These values adhere to the constraint specifications of 140, 100, and 180.

Slack Analysis: The slack values are identified in the report with the `.slack` suffix. They indicate the extent to which resources are underutilized in the optimal solution. Specifically, the CFC constraint is exceeded by $121.7778 - 100 = 21.7778$ thousand kg. In contrast, the CO2 and N2O constraints are met precisely, with zero slack (0 and -2.8422×10^{-14}). This suggests that the CFC constraint is non-binding, offering flexibility for adjustments without altering the optimal configuration.

[1] see Appendix, Part A on p.14 for output from .run file.

Shadow Price Analysis: The shadow prices are extracted from the report using the `.dual` suffix. They represent the marginal change in the objective function value for a one-unit alteration in the constraint's right-hand side, holding other factors constant (*ceteris paribus*). For CO2 and N2O, the shadow prices are 80 and 10, respectively. This indicates that increasing the reduction targets for CO2 and N2O by one unit would add 80 and 10 to the total cost. On the other hand, the shadow price for CFC is 0, signifying that changes in the CFC reduction target will not influence the total cost. This observation aligns with the slack value for CFC, corroborating its non-binding status in the current optimization model. The adjustable range is 90 to 144 for G1, 0 to 121.778 for G2, and 175 to 280 for G3.

Part A Task 3

How sensitive is the optimal cost to the targets set by the government?

Shadow prices, identified using the `.dual` suffix, are 80 for CO2, 0 for CFC, and 10 for N2O. These values indicate that the optimal cost is most sensitive to adjustments in the CO2 target. Specifically, a unit increase in the reduction targets for CO2, CFC, and N2O would increase the total cost by 80, 0, and 10 units, respectively, assuming all other factors remain constant.

If the target of N2O change to 155/220 thousand kg, what is the effect?

Utilizing the `cplex_options 'sensitivity'` option and the `.current`, `.down`, and `.up` suffixes, we determined that the allowable range for the shadow price is $[175, 280]$. A reduction to 155 thousand kg falls outside this range, implying that such a change could have broader implications on the model beyond the constraints on G3. Within the range down to 175 thousand kg, the shadow price remains valid, and we can infer that the optimal cost would decrease by $10 \times (180 - 175) = 50$ within this interval. For the range from 175 to 155, no conclusions can be made without rerunning the model.

Conversely, if the N2O target increases to 220 thousand kg, within the allowable range of $[175, 280]$, the optimal cost would rise by $10 \times (220 - 180) = 400$. Assuming all other variables remain constant, this would result in a new optimal cost of $13,000 + 400 = \$13,400$. In this case, we can definitively assess the impact on the optimal price.

Part A Task 4

How sensitive are the decisions to the accuracy of these coefficients?

Utilizing the `.down`, and `.up` suffixes, we ascertain the permissible ranges for fluctuations in the cost coefficients. Specifically, the ranges are $[462.5, 740]$ for SCR, $[1290, \infty]$ for LEF, and $[750, 1200]$ for EP. The current values for these coefficients are 500, 3300, and 1110, respectively. The reduced costs for SCR, LEF, and EP are 0, 2010, and 0, which aligns with the expectation since SCR and EP fall within their allowable ranges. Consequently, any decrease in the cost coefficients for SCR and EP would immediately yield a new optimal solution. For LEF, a reduction in its cost coefficient by 2010 would render it part of the optimal solution, while an increase would continue to exclude it, assuming all else remains constant (*ceteris paribus*).

If cost of LEF change from 3300 to 2950, would the optimal decisions change?

Should the cost coefficient for LEF decrease from 3300 to 2950, the reduced cost is expected to decline but remain positive. Specifically, it would be 1660 (i.e., $3300 - 2950 + 2010$). This implies that incorporating LEF into the optimal solution would still not offer a cost advantage over the other methods. As a result, the optimal decisions are likely to remain unchanged, with minimal impact on the overall optimal cost.

PART B

Part B Task 1 - Formulate a linear programming model in AMPL

Sets

C : Set of Cardboard P : Set of Products S : Set of Suppliers

Parameters

$$
\begin{aligned}
\text{purchase\_price}(c,s) &: \forall c \in C, s \in S : \text{Price of purchasing cardboard } c \text{ from source } s \\
\text{max\_supply}(c,s) &: \forall c \in C, s \in S : \text{Maximum supply of cardboard } c \text{ from source } s \\
\text{prod\_cost}(p) &: \forall p \in P : \text{Production cost of product } p \\
\text{sales\_price}(p) &: \forall p \in P : \text{Sales price of product } p \\
\text{prod\_capacity}(p) &: \forall p \in P : \text{Production capacity for product } p \\
\text{min\_demand}(p) &: \forall p \in P : \text{Minimum demand for product } p \\
\text{strength}(c) &: \forall c \in C : \text{Cardboard } c\text{'s strength} \\
\text{transparency}(c) &: \forall c \in C : \text{Cardboard } c\text{'s transparency} \\
\text{min\_strength}(p) &: \forall p \in P : \text{Minimum required strength for product } p \\
\text{max\_transparency}(p) &: \forall p \in P : \text{Maximum allowed transparency for product } p \\
A(p) &: \forall p \in P : \text{Binary parameter associated with product } p \\
B(p) &: \forall p \in P : \text{Another binary parameter associated with product } p
\end{aligned}
$$

Variables

$$
\begin{aligned}
x(c,p) \geq 0 &: \forall c \in C, p \in P : \text{Amount of cardboard } c \text{ used for product } p \\
y(c,s) \geq 0 &: \forall c \in C, s \in S : \text{Amount of cardboard } c \text{ purchased from source } s
\end{aligned}
$$

Objective:

$$
\text{Maximize total\_profit} =
$$

$$
\sum_{c \in C, p \in P} \text{sales\_price}(p) \times x(c,p) - \sum_{p \in P} \text{prod\_cost}(p) \times \sum_{c \in C} x(c,p) \quad - \sum_{c \in C, s \in S} \text{purchase\_price}(c,s) \times y(c,s)
$$

Constraints

$$
\begin{aligned}
\text{cardboard\_supply}(c,s) &: \forall c \in C, s \in S : y(c,s) \leq \text{max\_supply}(c,s) \\
\text{cardboard\_used}(c) &: \forall c \in C : \sum_{p \in P} x(c,p) \leq \sum_{s \in S} y(c,s) \\
\text{production\_capacity}(p) &: \forall p \in P : \sum_{c \in C} x(c,p) \leq \text{prod\_capacity}(p) \\
\text{minimum\_demand}(p) &: \forall p \in P : \sum_{c \in C} x(c,p) \geq \text{min\_demand}(p) \\
\text{strength\_constraint}(p) &: \forall p \in P : \sum_{c \in C} x(c,p) \times \text{strength}(c) \geq \text{min\_strength}(p) \times \sum_{c \in C} x(c,p) \\
\text{transparency\_constraint}(p) &: \forall p \in P : \sum_{c \in C} x(c,p) \times \text{transparency}(c) \leq \text{max\_transparency}(p) \times \sum_{c \in C} x(c,p) \\
\text{minimum\_purchase}(s) &: \forall s \in S : \sum_{c \in C} \text{purchase\_price}(c,s) \times y(c,s) \geq 1500000
\end{aligned}
$$

$$
\text{max\_A} : \sum_{c \in C, p \in P} x(c,p) \times A(p) \leq 50000, \quad \text{max\_B} : \sum_{c \in C, p \in P} x(c,p) \times B(p) \leq 45000
$$

What is the optimal blending plan and how much profit does it provide?

The optimal blending plan and the associated profit are depicted in the table below. The model yields a maximum profit of $2,915,520. All numerical values in Part B, task 1 are sourced from the `B.txt` file, generated by executing the `B.run` file. The complete output is provided in the Appendix for reference[2].

C1 S1	C1 S2	C2 S1	C2 S2	C3 S1	C3 S2	C4 S1	C4 S2
0	20000	25955.7	0	0	23132.9	911.392	20000

C1 P1	C1 P2	C1 P3	C1 P4	C2 P1	C2 P2	C2 P3	C2 P4	C3 P1	C3 P2	C3 P3	C3 P4	C4 P1	C4 P2	C4 P3	C4 P4
0	0	5000	15000	0	5955.7	5000	15000	0	23132.9	0	0	9000	11911.4	0	0

Which supplier is most profitable?

To gauge supplier profitability, we consider several metrics. Initially, it's reasonable to assume that the objective function, aimed at maximizing profit, would naturally favor the most profitable supplier. **Slack analysis:** We examine the slack associated with the `minimum_purchase` constraint, as indicated by the `.slack` suffix in the B.run file. See the output text file in the appendix under the "Part B" section. The slack values for suppliers S1 and S2 are 0 and $1,864,480$, respectively. This suggests that supplier S2 is more profitable. Specifically, we exceed the minimum purchase requirement from S2 by a total value of $1,864,480$ while meeting only the bare minimum for S1.

Part B Task 2a

To determine the maximum price per tonne that the company should consider from a new supplier, we aim to achieve at least a 2.5% increase in optimal profit compared to the baseline established in Task 1. Under the assumption that all other parameters and conditions remain unchanged, we refer to the shadow price of constraint $C1$, which stands at 134.177. This value is obtained from the `.dual` suffix in the output of Task 1, as detailed in the appendix under Part B. Additionally, the output reveals an allowable range of 1285.71 for the shadow price, confirming its validity for an increase of 1000 units.

We formulate an equation where the unknown variable represents this cost to identify the maximum acceptable price per tonne `max_price`. The equation ensures that the profit increase meets or exceeds 2.5% compared to the optimal profit from Task 1. Solving this equation will yield the maximum price per tonne that the company should consider for purchasing 1000 tonnes from the new supplier which is **61.289**.

$$2915520 + 1000 \times (134.177 - \text{max\_price}) = 2915520 \times (1.025)$$
$$\text{max\_price} = 61.289$$

[2]see Appendix, Part B, task 1 on p.15 for output from .run file.

Part B Task 2b

Considering the introduction of cardboard type C5, it is characterized by a strength value of 97 and a transparency level of 0.3%. These attributes place C5 in an intermediary position between cardboard types C1 and C2, both in strength and transparency.

Cardboard	Strength	Transparency	Shadow price
C1	96	0.2%	134.177
C5	97	0.3%	NA
C2	98	0.4%	55.8228

Utilizing the principle of "linearity of yield," we can infer that the shadow price of C5 would likely be the arithmetic mean of the shadow prices for C1 and C2. Mathematically, this can be expressed as:

$$\text{Estimated Shadow Price of C5} = \frac{(134.177 + 55.8228)}{2} = 95$$

Maximum price the company should accept to pay per each tonne

To identify the maximum price the company should consider for C5, we formulate an equation where x, the unknown price, is the variable to solve for. This equation aims to balance two critical factors: the left-hand side represents the company's 2.5% profit increase target, calculated as $0.025 \times 2,915,520$, while the right-hand side captures the added value from acquiring 1000 tonnes of C5 at a price x below its estimated shadow price of 95.

$$0.025 \times 2915520 = 1000 \times (95 - x)$$
$$x = 22.112$$

Thus, the maximum price the company should be willing to pay for C5 is approximately **$22.11 per tonne**. To corroborate our analytical findings, we extended our AMPL model to accommodate the new scenario. The salient modifications to the model are outlined below and can be further examined in the `B2b.dat`, and `B2b.mod` files:

Data File Augmentation: The `.dat` file was updated to incorporate a new supplier, labeled as S3. This update required adjustments to several parameters, including `max_supply`, `purchase_price`, and the strength and transparency attributes of the new cardboard type. **Constraint Refinement:** The `minimum_purchase` constraint is refined to omit the newly introduced supplier, S3 specifically. This is accomplished by adjusting the summation index to $\{s \in S : s \neq \text{'S3'}\}$. **Incorporation of Binary Variable:** A binary variable z is introduced to ensure that either 0 or 1000 units of cardboard C5 will be purchased from S3. With this variable, we can formulate a new constraint to link the purchased quantity of cardboard C5 from S3 to the binary variable z

```
s.t. cardboard_supply_S3: y['C5','S3'] = 1000 * z;
```

Upon running the modified model, we obtained a shadow price of 95 for cardboard type C5, which aligns with our initial assumption. The optimized profit reached approximately $2,988,410, a 2.5% increase compared to the original model. This result, which can be found in appendix [3], empirically validates our initial hypothesis regarding the shadow price of C5.

[3] see Appendix, Part B, Task 2 on p.16 for output from .run file.

PART C

Part C Task 1 - Formulate a linear programming model

We formulate the following model to minimize the monthly transport costs in meeting demands of all markets.

Sets

R : Set of regions $\quad$ F : Set of facilities $\quad$ K : Set of markets

Parameters

$$\text{max\_supply}(r) : \forall r \in R : \text{Maximum supply from region } r$$
$$\text{max\_prod}(f) : \forall f \in F : \text{Maximum production capacity of facility } f$$
$$\text{region\_to\_facility}(r, f) : \forall r \in R, f \in F : \text{Transport cost from region } r \text{ to facility } f$$
$$\text{facility\_to\_market}(f, k) : \forall f \in F, k \in K : \text{Transport cost from facility } f \text{ to market } k$$
$$\text{demand}(k) : \forall k \in K : \text{Demand at market } k$$

Decision Variables

$$x(r, f) \geq 0 : \forall r \in R, f \in F : \text{Quantity transported from region } r \text{ to facility } f$$
$$y(f, k) \geq 0 : \forall f \in F, k \in K : \text{Quantity transported from facility } f \text{ to market } k$$

Objective:

$$\text{Minimize transport\_cost} =$$
$$\sum_{r \in R, f \in F} \text{region\_to\_facility}(r, f) \times x(r, f) + \sum_{f \in F, k \in K} \text{facility\_to\_market}(f, k) \times y(f, k)$$

Constraints

$$\text{supply\_constraint}(r) : \forall r \in R : \sum_{f \in F} x(r, f) \leq \text{max\_supply}(r)$$
$$\text{production\_constraint}(f) : \forall f \in F : \sum_{r \in R} x(r, f) \leq \text{max\_prod}(f)$$
$$\text{demand\_constraint}(k) : \forall k \in K : \sum_{f \in F} y(f, k) \geq \text{demand}(k)$$
$$\text{supply\_equals\_demand}(f) : \forall f \in F : \sum_{k \in K} y(f, k) \leq \sum_{r \in R} x(r, f)$$

Implement the model in AMPL and solve it: What is the optimal shipping plan and how much does it cost?

The optimal shipping plan from regions to facilities and from facilities to markets are shown in the two tables below based on the output of C.run file, see appendix [4]. The total cost to transport goods and meet market demands is $10,480.

Region r	R1	R1	R2	R2
Facility f	F1	F2	F1	F2
Quantity x(r,f)	150	0	0	137

$F1$ is solely supplied by $R1$ and $F2$ by $R2$. Markets receive salmon from only one facility. Both facilities operate near their maximum capacity. The model successfully minimises transportation costs while meeting all constraints.

[4] see Appendix, PART C, task 1 on p.17 for output from .run file

Facility f	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1
Market k	K1	K10	K11	K12	K13	K14	K15	K2	K3	K4	K5	K6	K7	K8	K9
Quantity y(f,k)	0	28	0	18	0	15	23	21	15	15	0	0	0	0	15
Facility f	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2
Market k	K1	K10	K11	K12	K13	K14	K15	K2	K3	K4	K5	K6	K7	K8	K9
Quantity y(f,k)	17	0	19	0	25	7	0	0	0	0	22	15	20	12	0

Part C Task 2

Run your model for this new scenario.

In the revised scenario, the model anticipates a 25% increase in demand across all markets. This escalation amplifies the demand and considerably restricts the feasible solution space for achieving an optimal outcome. To accommodate this shift, the demand constraint within the model is adjusted to encapsulate the 25% uptick in market demand. I made the change in the `C2a.mod` file.

$$\text{demand\_constraint}(k) \quad \forall k \in K : \sum_{f \in F} y(f,k) \geq 1,25 \times \text{demand}(k)$$

What do you obtain?

When executed with these amended parameters, the model returns an `"infeasible"` status. This result is not unexpected, given the near-saturation of production capacities with only 13 units slack in Facility 2; see Appendix, Part C, Task 2a[5].

With the demand now inflated by 25%, this residual capacity is swiftly exhausted, rendering the problem infeasible and thereby confirming the limitations of the existing production infrastructure. The equation below highlights the issue of excessive expansion in production capacity, underlining the output in the appendix, task C2a.

$$\sum_{k=1}^{15} K_k = 17 + 21 + 15 + 15 + 22 + 15 + 20 + 12 + 15 + 28 + 19 + 18 + 25 + 22 + 23 = 287 < 300$$

$$1.25 \cdot \sum_{k=1}^{15} K_k = 358.75 > 300$$

Formulate a linear programming model, allowing for unsatisfied demand, but at a certain penalty cost

The company has anticipated a 25% increase in demand from each market for the upcoming month while allowing for penalties for unsatisfied demand. We modify the variables to include the non-negative unsatisfied demand (z). In the objective function, the amount of unsatisfied demand z is timed with the cost per unit of unsatisfied demand from market k. Demand is still increased by 1.25 in the constraints, and the unsatisfied demand cannot exceed this demand per market. To end, the demand from a market k equals the quantity exported y to the market k plus the amount of unsatisfied demand z.

Decision Variables

$$z(k) \geq 0 \quad \forall k \in K : \text{Amount of unsatisfied demand from market } k$$

Objective:

[5] see Appendix, Task 2a on p.18 for output from .run file.

$$\text{Minimize transport\_cost} = \sum_{r\in R, f\in F} x(r,f) \times \text{region\_to\_facility}(r,f) + \sum_{f\in F, k\in K} y(f,k) \times \text{facility\_to\_market}(f,k) + \sum_{k\in K} \text{penalty}(k) \times z(k)$$

Constraints

$$\text{demand\_constraint}(k): \quad \forall k \in K : \sum_{f\in F} y(f,k) + z(k) \geq 1.25 \times \text{demand}(k)$$

$$\text{unsatisfied\_demand\_constraint}(k): \quad \forall k \in K : z(k) \leq 1.25 \times \text{demand}(k)$$

Implement the model in AMPL and solve it: How much is the optimal cost, and how much of this cost is due to transportation and how much due to penalties?

For output from .run file after solving the model, check appendix [6]. Solving the model, we find a shipping plan, found in the tables at the bottom of the page. Furthermore, we find the following overview of the unsatisfied market, using the new variable $z(k)$.

Market k	K1	K10	K11	K12	K13	K14	K15	K2	K3	K4	K5	K6	K7	K8	K9
Unsatisfied z(k)	0	0	23.75	22.5	31.25	27.5	28.75	0	0	0	0	0	0	0	0
Penalty	100	50	20	20	20	20	0	100	100	100	50	50	50	50	50

From this, we can calculate the following transport cost from region to facility and facility to market, but also the total penalty cost from the unsatisfied markets. Total transport cost from region to facility: $117.5 \cdot 13 + 107.5 \cdot 11 = \$2710$, total transport cost from facility to market: $35 \cdot 24 + 26.25 \cdot 9 + 18.75 \cdot 9 + 18.75 \cdot 21 + 18.75 \cdot 18 + 21.25 \cdot 15 + 27.5 \cdot 21 + 18.75 \cdot 24 + 25 \cdot 27 + 15 \cdot 87 = \$4312.5$, and total penalty cost: $23.75 \cdot 20 + 22.5 \cdot 20 + 31.25 \cdot 20 + 27.5 \cdot 20 = \$2100$. The optimal cost is found from the objective function, but is also of course the sum of transport and penalty costs: $\$2710 + \$4312.5 + \$2100 = \$9,122.5$.

How much is the unsatisfied demand for this market, and which market is the one with the largest amount of unsatisfied demand?

Total unsatisfied demand is $23.75 + 22.5 + 31.25 + 27.5 + 28.75 = 133.75$ tonnes, where market $K13$ has the largest unsatisfied demand, with 31.25 tonnes.

Is the total capacity of the facilities used at its maximum?

$F1$ is operating at 117.5 tonnes, below its maximum capacity of 150 tonnes. $F2$ is operating at 107.5 tonnes, also below its maximum capacity of 150 tonnes.

Region r	R1	R1	R2	R2
Facility f	F1	F2	F1	F2
x(r,f)	117.5	0	0	107.5

Facility f	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1
Market k	K1	K10	K11	K12	K13	K14	K15	K2	K3	K4	K5	K6	K7	K8	K9
y(f,k)	0	35	0	0	0	0	0	26.25	18.75	18.75	0	0	0	0	18.75
Facility f	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2	F2
Market k	K1	K10	K11	K12	K13	K14	K15	K2	K3	K4	K5	K6	K7	K8	K9
y(f,k)	21.25	0	0	0	0	0	0	0	0	0	27.5	18.75	25	15	0

[6] see Appendix, Task C2b on p.20 for txt output from .run file.

PART D

Part D Task 1a - Infeasibility

This model can be infeasible if the data provided cannot satisfy all constraints simultaneously. Several scenarios could make the model infeasible. Here are a couple of key examples:

Insufficient Charging Window: Our interpretation of constraint (2) suggests that users can specify s_{end} as they wish. However, if the available charging time window $[i, f]$ for a given vehicle k is insufficient to reach the desired state-of-charge s_{end}, the model becomes infeasible, violating equation (2). Infeasibility may arise as the upper limit of $x(k, t)$ is constrained by equations (3) and (4), making it unattainable to meet the state-of-charge requirements at both the start and end of the charging period.

Exceeding Maximum Capacity: If the sum of the power charged by all cars $x_{k,t}$ at any time t exceeds the maximum charging capacity c_t, the model would be infeasible as it would violate the constraint on the total fleet charging capacity. The total charging capacity (c_t) might be too low to accommodate the charging needs of all vehicles, especially if many vehicles need to be charged simultaneously.

Part D Task 1b

Is the model unbounded if all parameters are non-negative?

In the context of this optimization model, the concept of unboundedness would imply that the objective function, designed to minimize the total cost of charging vehicles, could theoretically decrease without limit, potentially reaching negative infinity. However, this scenario is unlikely to occur for the following reasons: **1:** Given the assumption that all parameters are non-negative, this naturally extends to the price per kWh p_t also being non-negative. **2:** The objective function is formulated as $\min \sum_{k \in K} \sum_{t \in T} p_t \cdot x_{k,t}$. Since both p_t and $x_{k,t}$ are non-negative, their product will also be non-negative. This establishes a lower bound of 0 for the objective function. Therefore, the model is unlikely to be unbounded under the given conditions.

Part D Task 1c

Is model unbounded if all parameters except for price is non-negative?

In this optimisation model, the objective function aims to minimise the total charging cost. If the price per kWh (p_t) is negative for some time, the model could become unbounded. This is because the objective function is the product of the amount of energy charged (x) and the price (p) at each time period.

1: If p were to be negative infinity, multiplying it by any x would result in an infinitely negative objective function value. Since there is no explicit lower bound on how negative p can be, this poses a risk of the model becoming unbounded. **2:** Additionally, constraints (3) and (4) set upper bounds $m(k)$ and $c(t)$ for the charging rates. If these constraints were to be put to positive infinity, then x could also be infinitely large, further exacerbating the risk of an unbounded objective function when p is any negative number. **In conclusion:** The model has the potential to be unbounded under specific conditions: either when p is infinitely negative and x is any non-negative number, or when p is any negative number, and x is infinitely positive. This is due to the absence of an upper bound on constraints (3) and (4) and no lower bound on the price.

Part D Task 2

Update the linear model to account for this new condition

We already know the total power consumption in period t: power_consumption(t) $\quad \forall t \in T, k \in K : \sum_{k \in \mathcal{K}} x_{k,t}$.

After that, we can add a constraint on increasing or decreasing the amount charged for all t in T such that $t + 1$ also belongs to T. The critical part here is that the absolute value does not vary too much from the previous value, which is captured by adding one constraint for increase and one for decrease. One might wonder why we don't use an absolute value to capture the difference, as in $|x_{k,t+1} - x_{k,t}| \leq \delta$. The reason is that absolute value functions introduce non-linearity into the model. We can capture the same condition using two separate constraints while keeping the model linear.

$$\text{increase\_constraint}(t): \quad \sum_{k \in \mathcal{K}} x_{k,t+1} - \sum_{k \in \mathcal{K}} x_{k,t} \leq \delta, \quad \text{decrease\_constraint}(t): \quad \sum_{k \in \mathcal{K}} x_{k,t} - \sum_{k \in \mathcal{K}} x_{k,t+1} \leq \delta$$

Consequences of this modification compared to original model

Adding any constraint will inherently alter the set of feasible solutions, potentially making the model less adaptable to various scenarios.**1: Reduced Flexibility:** The new constraints limit the model's ability to adapt to different scenarios and could make it less optimal in cost minimization. **2: Potential Infeasibility:** In some instances, mainly when δ is set to a low value, ensuring that the difference in power consumption between any two consecutive hours does not exceed δ could be challenging while respecting other constraints in the model. This could lead to an infeasible solution. **3: Smoothing Effect:** On the positive side, the smoothing condition could lead to a more balanced utilization of the energy grid, reducing spikes in demand and potentially lowering overall energy costs in a real-world scenario.

Part D Task 3a

Can you conclude how v^* compares with respect to 37,500?

The objective of the model is to minimize the total cost of charging. Given this, the model is likely to adapt to the new prices at time periods $T = 12$ and $T = 20$ by shifting the charging times, provided that the constraints allow for such flexibility. A price hike at $T = 12$ would generally deter charging during this period, while a price drop at $T = 20$ would likely encourage more charging, all within the bounds of existing constraints like battery capacity and charging station availability. **1: Constraints Limitations**: The model's adaptability is not unlimited and is constrained by various factors. These could include state-of-charge requirements, charging station availability, and vehicle-specific constraints. Without these details, it's challenging to precisely predict the impact on v^*. **2: Demand Variability**: If demand around $T = 12$ is substantially higher than at $T = 20$, the model may find it difficult to redistribute all the demand to adjacent, cheaper time slots without violating other constraints, potentially leading to a less optimal v^*. **3: Net Effect on Objective Value**: While the price increase at $T = 12$ is counterbalanced by the decrease at $T = 20$, the net effect on v^* is not guaranteed to be neutral. This is due to varying demand levels, the model's constrained flexibility, and how much of the total demand can be shifted to take advantage of the lower price at $T = 20$.

In summary, the new v^* could be better, the same, or worse than the original 37,500 NOK, depending on a multitude of factors such as demand variability, constraint limitations, and the model's inherent sensitivity to price changes.

Part D Task 3b

How does w^* compare with 37,500?

1: Increased Flexibility: The extended charging window for "TeslaX" allows the model more options for when to charge this car. This could enable the model to capitalize on lower electricity prices during this period, potentially reducing the overall cost. **2: Unchanged State-of-Charge Requirement:** The desired state-of-charge for "TeslaX" remains the same, meaning the car doesn't require more electricity, just more time to achieve the same charge level. **3: No New Constraints:** Given that no other data has changed, the model faces no additional constraints that could increase the cost. However, it's worth noting that this is an assumption based on the problem statement. **4: Market Demand and Capacity:** While "TeslaX" has more flexibility, it's important to consider how this could affect the overall demand and capacity. If many users have similar flexibility, the model may need to balance this against other constraints.

Summary: Based on these considerations, it is likely that w^* will be less than or equal to 37,500 NOK. The extended charging window for "TeslaX" offers more opportunities for cost minimization, making it improbable for w^* to exceed the original cost of 37,500 NOK.

Appendix

PART A Task 1-4

This is the `A.run` file printout (A.txt) from the `A.mod` file using data from the `A.dat` file.

```
total_cost = 13000

x [*] :=
M1   1.33333
M2   0
M3  11.1111
;

:  reduction_emissions.body reduction_emissions.dual    :=
G1          140                          80
G2          121.778                       0
G3          180                          10
;

reduction_emissions.slack [*] :=
G1  -2.84217e-14
G2  21.7778
G3   0
;

:       x       x.rc   x.down   x.up     :=
M1    1.33333      0    462.5     740
M2    0         2010   1290     1e+20
M3   11.1111       0    750      1200
;

:  reduction_emissions.down reduction_emissions.up    :=
G1               90                 144
G2           -1e+20                 121.778
G3              175                 280
;
```

PART B

Task 1

This is the `B.run` file printout (B.txt) from the `B.mod` file using data from the `B.dat` file.

```
total_profit = 2915520

:          x          y        :=
C1 P1       0          .
C1 P2       0          .
C1 P3    5000          .
C1 P4   15000          .
C1 S1       .          0
C1 S2       .      20000
C2 P1       0          .
C2 P2    5955.7        .
C2 P3    5000          .
C2 P4   15000          .
C2 S1       .      25955.7
C2 S2       .          0
C3 P1       0          .
C3 P2   23132.9        .
C3 P3       0          .
C3 P4       0          .
C3 S1       .          0
C3 S2       .      23132.9
C4 P1    9000          .
C4 P2   11911.4        .
C4 P3       0          .
C4 P4       0          .
C4 S1       .        911.392
C4 S2       .      20000
;

:  minimum_purchase.body minimum_purchase.slack    :=
S1        1500000              -2.32831e-10
S2        3364480          1864480
;

cardboard_used['C1'].dual = 134.177
cardboard_used['C1'].up = 1285.71
```

Task 2

This is the `B2b.run` file printout (B2b.txt) from the `B2b.mod` file using data from the `B2bdat` file.

```
total_profit = 2988410

cardboard_used['C5'].dual = 95
```

PART C

Task 1

This is the `C.run` file printout (C.txt) from the `C.mod` file using data from the `X.dat` file.

```
transport_cost = 10480
:          x    y     :=
F1 K1      .     0
F1 K10     .    28
F1 K11     .     0
F1 K12     .    18
F1 K13     .     0
F1 K14     .    15
F1 K15     .    23
F1 K2      .    21
F1 K3      .    15
F1 K4      .    15
F1 K5      .     0
F1 K6      .     0
F1 K7      .     0
F1 K8      .     0
F1 K9      .    15
F2 K1      .    17
F2 K10     .     0
F2 K11     .    19
F2 K12     .     0
F2 K13     .    25
F2 K14     .     7
F2 K15     .     0
F2 K2      .     0
F2 K3      .     0
F2 K4      .     0
F2 K5      .    22
F2 K6      .    15
F2 K7      .    20
F2 K8      .    12
F2 K9      .     0
R1 F1    150    .
R1 F2      0    .
R2 F1      0    .
R2 F2    137    .
;
```

Task 2a

This is the `C2a.run` file printout (C2a.txt) from the `C2a.mod`, `C.mod` files using data from the `C.dat` file.

```
Allowable range under optimal solution in question 1:

:  production_constraint.body production_constraint.slack    :=
F1             150                             0
F2             137                            13
;

Constraint is not satisfied when increasing demand

:  production_constraint.body production_constraint.slack    :=
F1            150                             0
F2            208.75                        -58.75
;
```

Task 2b

This is the `C2b.run` file printout (C2b.txt) from the `C2b.mod` file using data from the `C2b.dat` file.

```
transport_cost = 9122.5

:          x       y      :=
F1 K1      .       0
F1 K10     .      35
F1 K11     .       0
F1 K12     .       0
F1 K13     .       0
F1 K14     .       0
F1 K15     .       0
F1 K2      .      26.25
F1 K3      .      18.75
F1 K4      .      18.75
F1 K5      .       0
F1 K6      .       0
F1 K7      .       0
F1 K8      .       0
F1 K9      .      18.75
F2 K1      .      21.25
F2 K10     .       0
F2 K11     .       0
F2 K12     .       0
F2 K13     .       0
F2 K14     .       0
F2 K15     .       0
F2 K2      .       0
F2 K3      .       0
F2 K4      .       0
F2 K5      .      27.5
F2 K6      .      18.75
F2 K7      .      25
F2 K8      .      15
F2 K9      .       0
R1 F1    117.5     .
R1 F2      0       .
R2 F1      0       .
R2 F2    107.5     .
;

z [*] :=
 K1   0
K10   0
K11  23.75
K12  22.5
K13  31.25
```

```
K14  27.5
K15  28.75
 K2   0
 K3   0
 K4   0
 K5   0
 K6   0
 K7   0
 K8   0
 K9   0
;

sum{r in R, f in F} x[r,f]*region_to_facility[r,f] = 2710

sum{ff in F, k in K} y[ff,k]*facility_to_market[ff,k] = 4312.5

sum{kk in K} penalty[kk]*z[kk] = 2100
```

PART D

Sets:

K	Set of vehicles
T	Set of time periods (from 1 to 24)
U	Set of tuples (k, i, f) such that vehicle k is available for charging during the time interval $[i, f]$

Parameters:

$s\_start(k, i)$ $\forall k \in K, i \in F$: State-of-charge of vehicle k at the beginning of period i

$s\_end(k, f)$ $\forall k \in K, i \in F$: State-of-charge of vehicle k desired at the end of period f

$m(k)$ $\forall k \in K$: Maximum charge per time period allowable for vehicle k

$p(t)$ $\forall t \in T$: Price per kWh at time t

$c(t)$ $\forall t \in T$: Maximum charging capacity utilisation by the total fleet of vehicles during time period t

Decision Variables:

$$x(k,t) \geq 0 \quad \forall k \in K, t \in T : \text{Power charged by vehicle } k \text{ during period } t$$

Objective:

$$\text{Minimize} \quad \text{total\_charging\_cost} = \sum_{k \in K} \sum_{t \in T} p(t) \cdot x(k,t)$$

Constraints:

$$\text{battery\_charge}(k,i,f) \quad \forall (k,i,f) \in U : s\_start(k,i) + \sum_{t=i}^{f} x(k,t) = s_{\text{end}}(k,f)$$

$$\text{max\_charge\_rate}(k,t) \quad \forall k \in K, \forall t \in T : x(k,t) \leq m(k)$$

$$\text{fleet\_capacity\_utilization}(t) \quad \forall t \in T : \sum_{k \in K} x(k,t) \leq c(t)$$